University of Engineering and Technology

2026-S-CYS-12

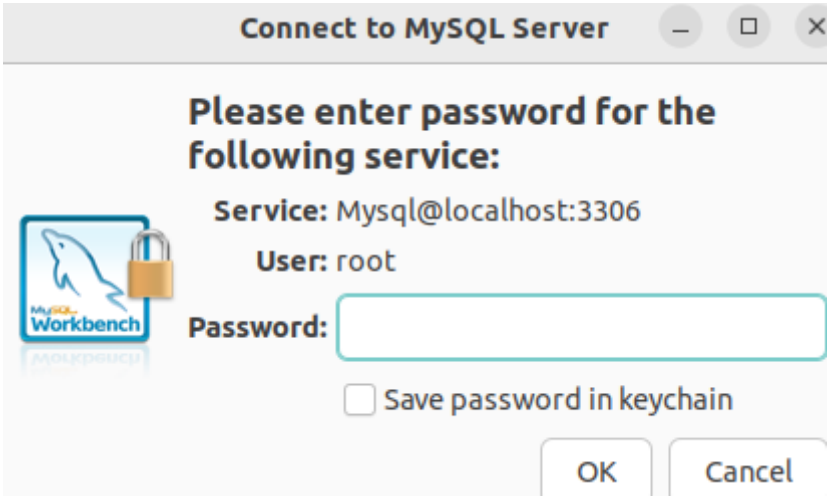
Anaya Azam

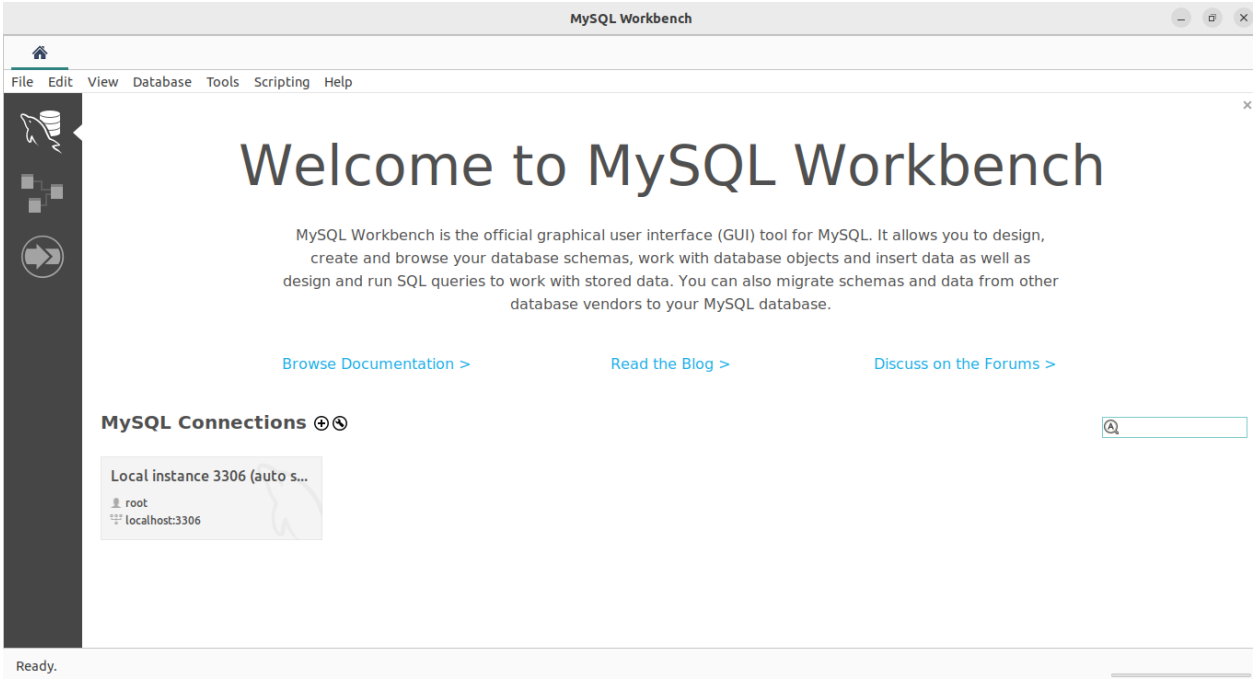
CMPE - 211L Database Systems

Lab – 01

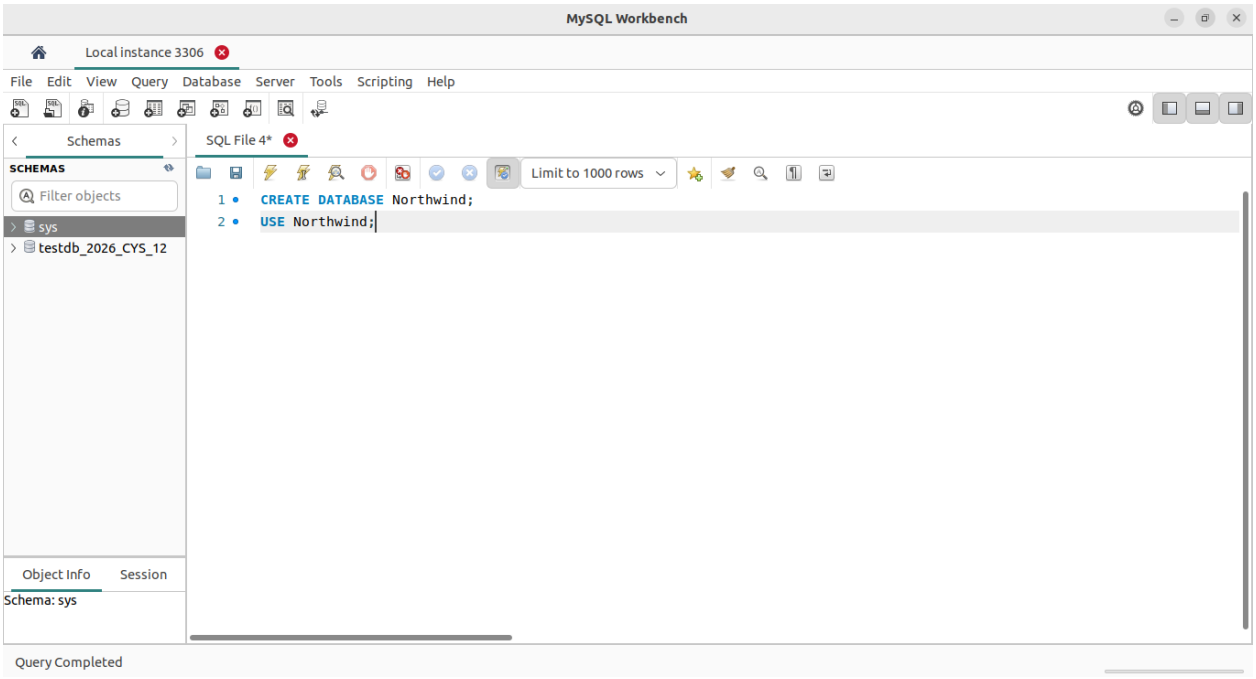
Due: 22nd September, 2026.

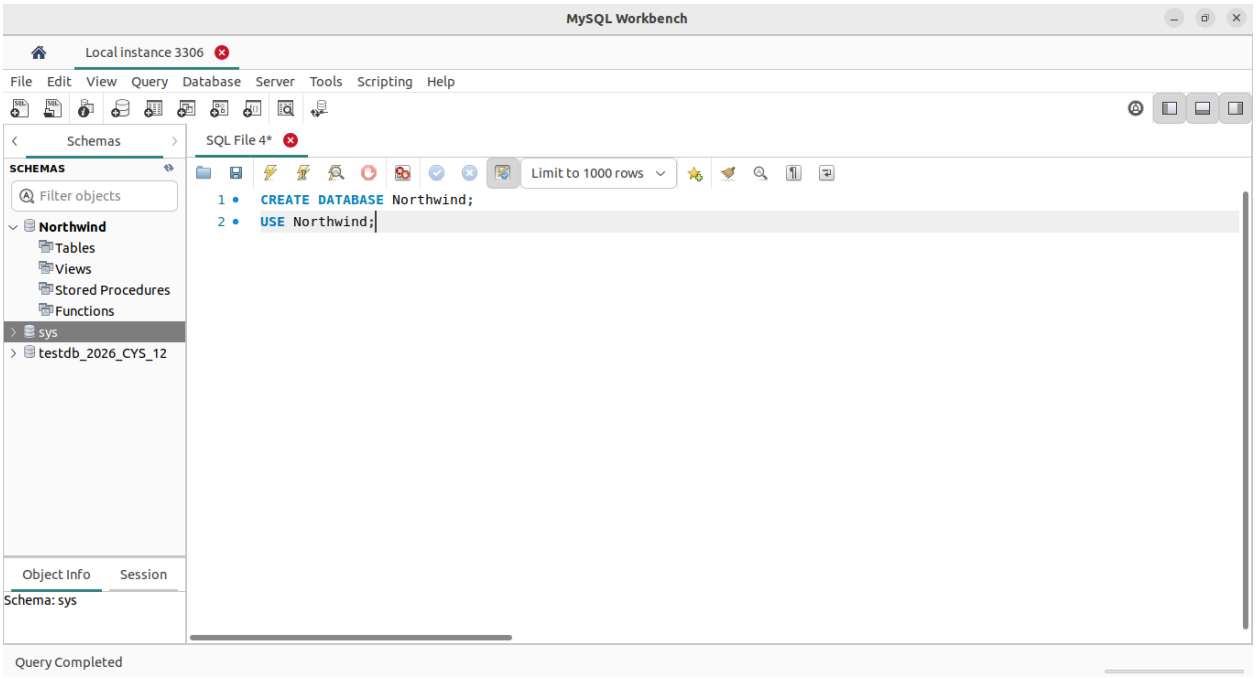
Learning Objective 1: Install and configure the MySQL server to prepare a working environment for database development and management.



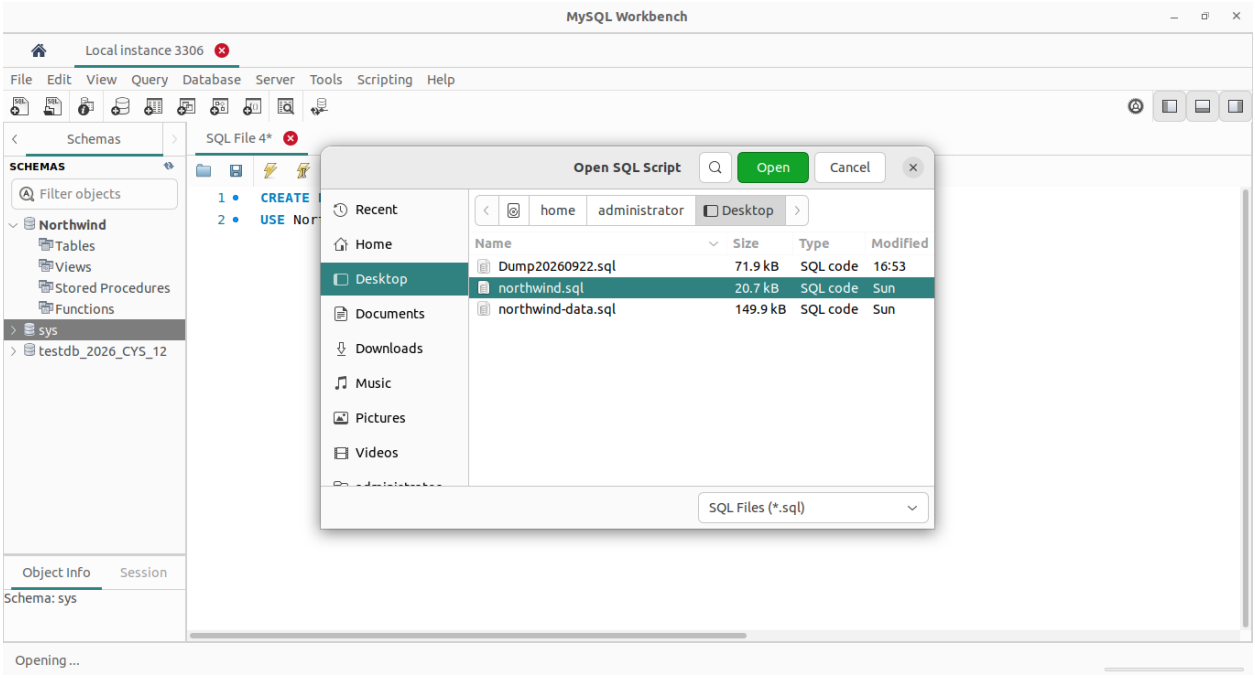


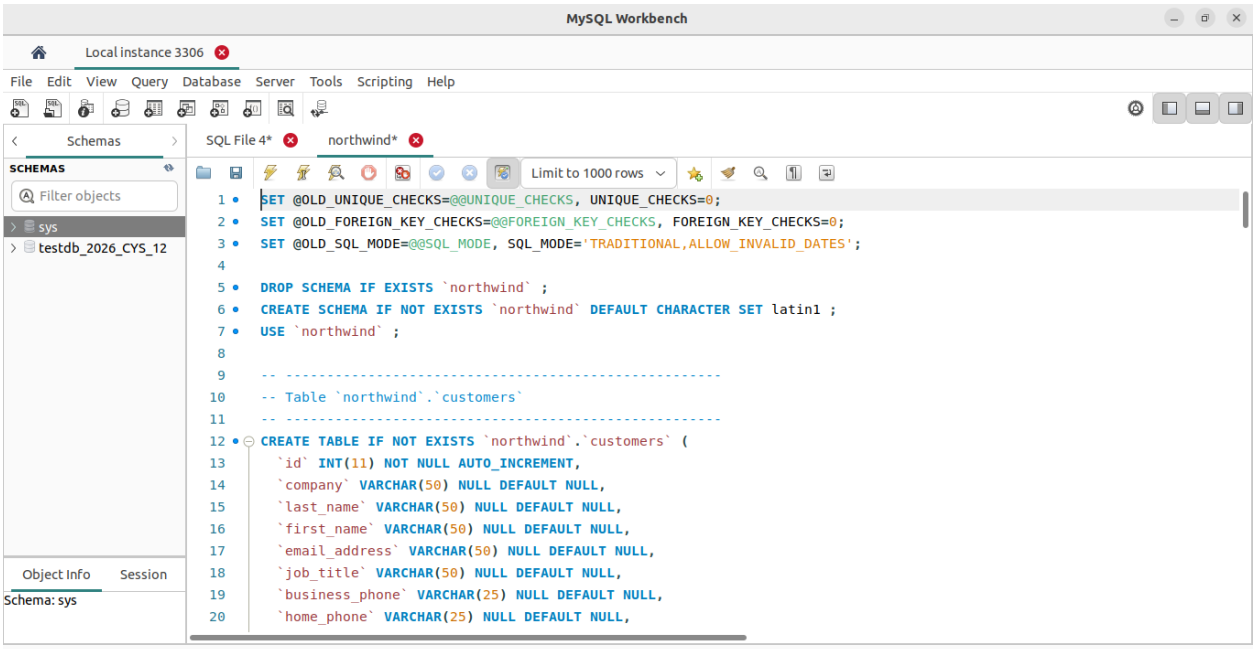
Learning Objective 2: Create a new database named "Northwind" to understand the basic process of setting up a Database.

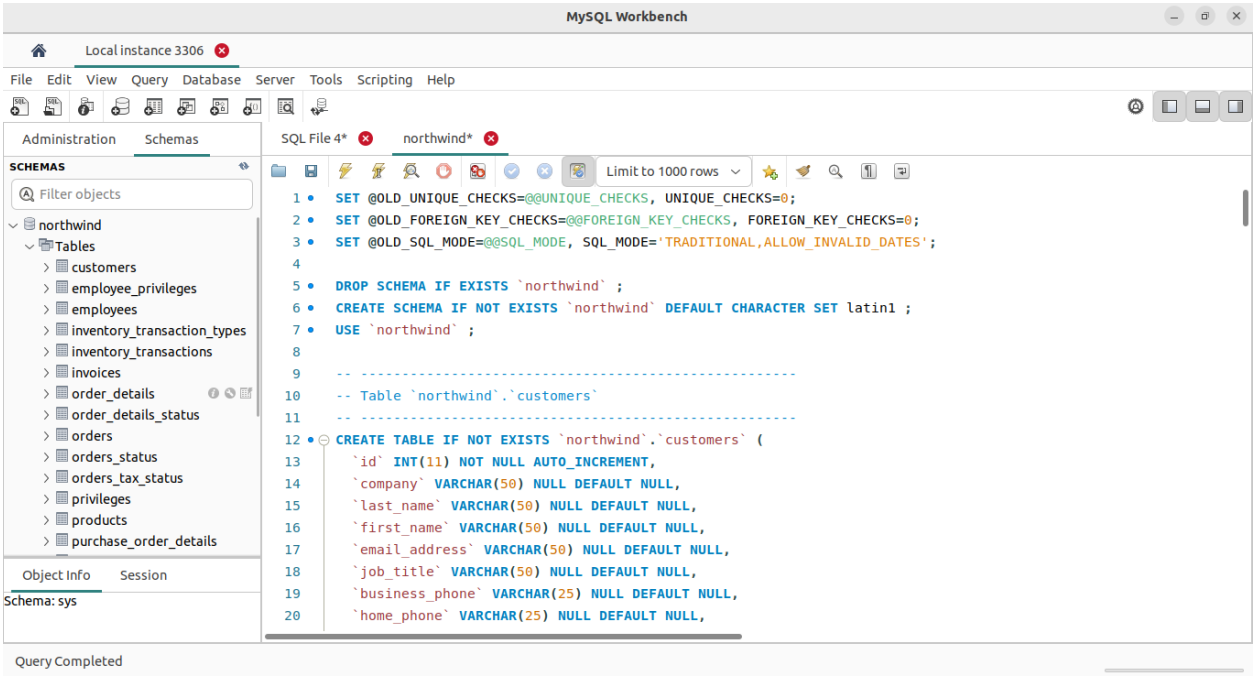




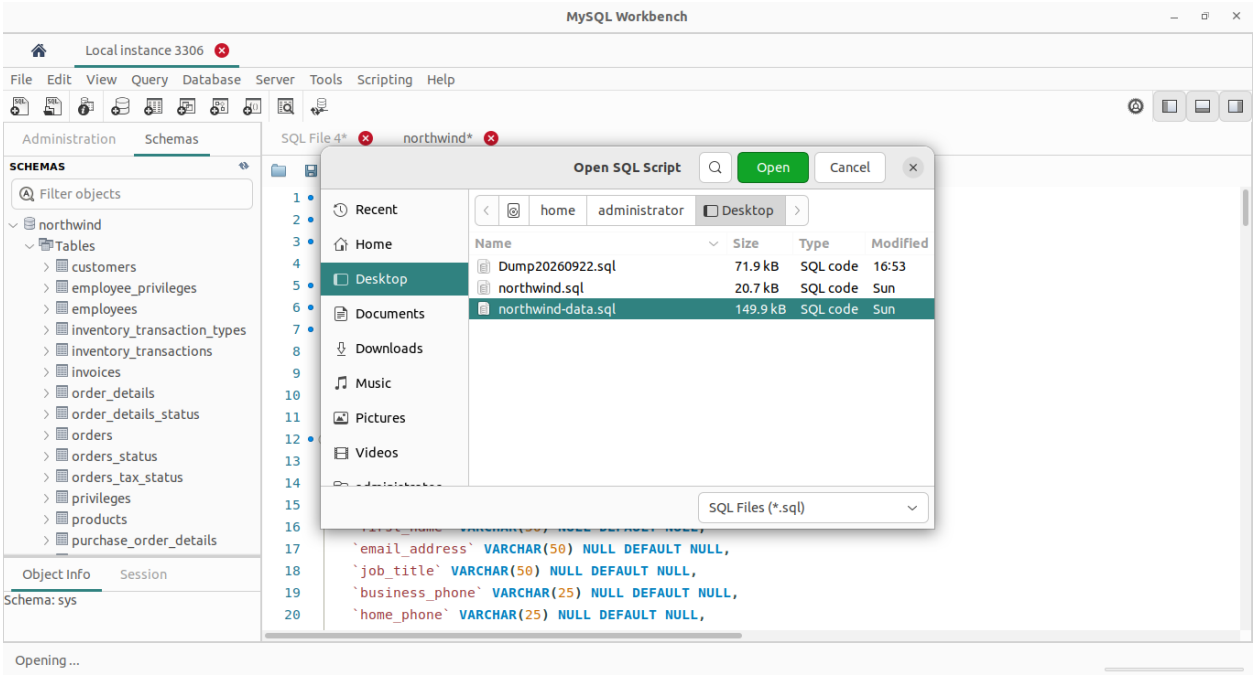
Learning Objective 3: Restore database schema by executing scripts (e.g., northwind.sql) to comprehend how schema scripts define database structure.

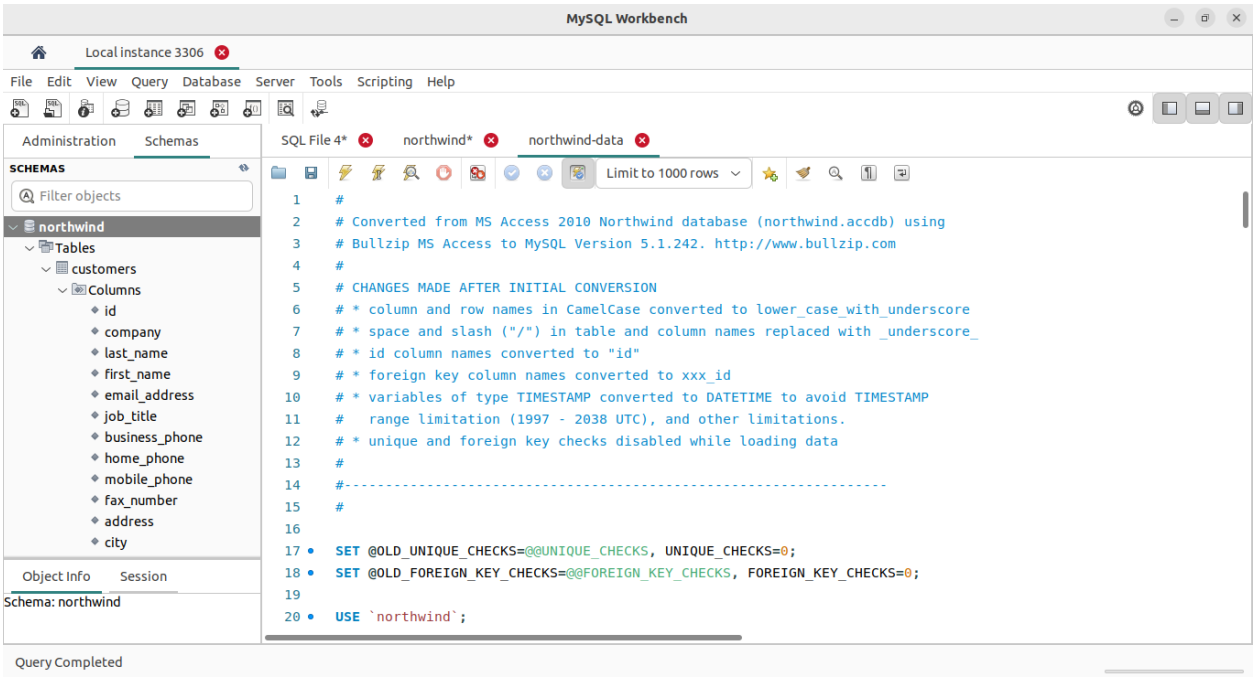


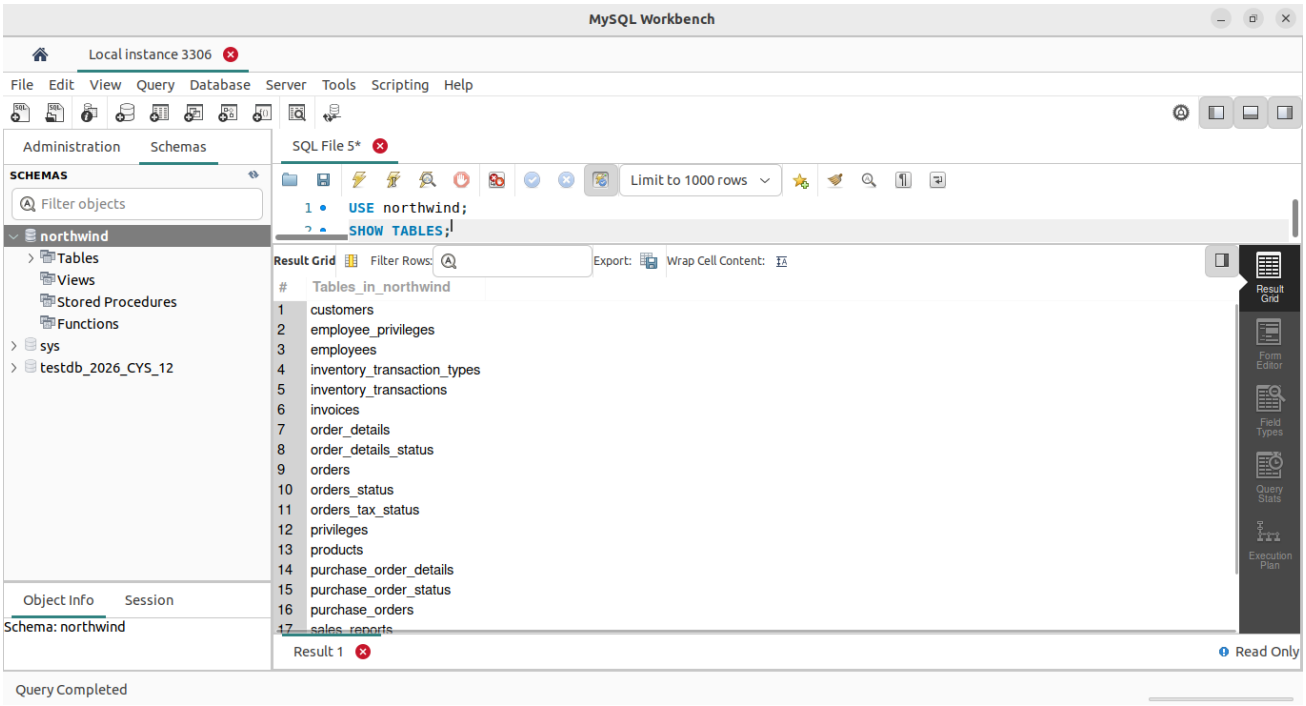




Learning Objective 4: Populate the database with data using northwind_data.sql to practice importing data into a structured database.

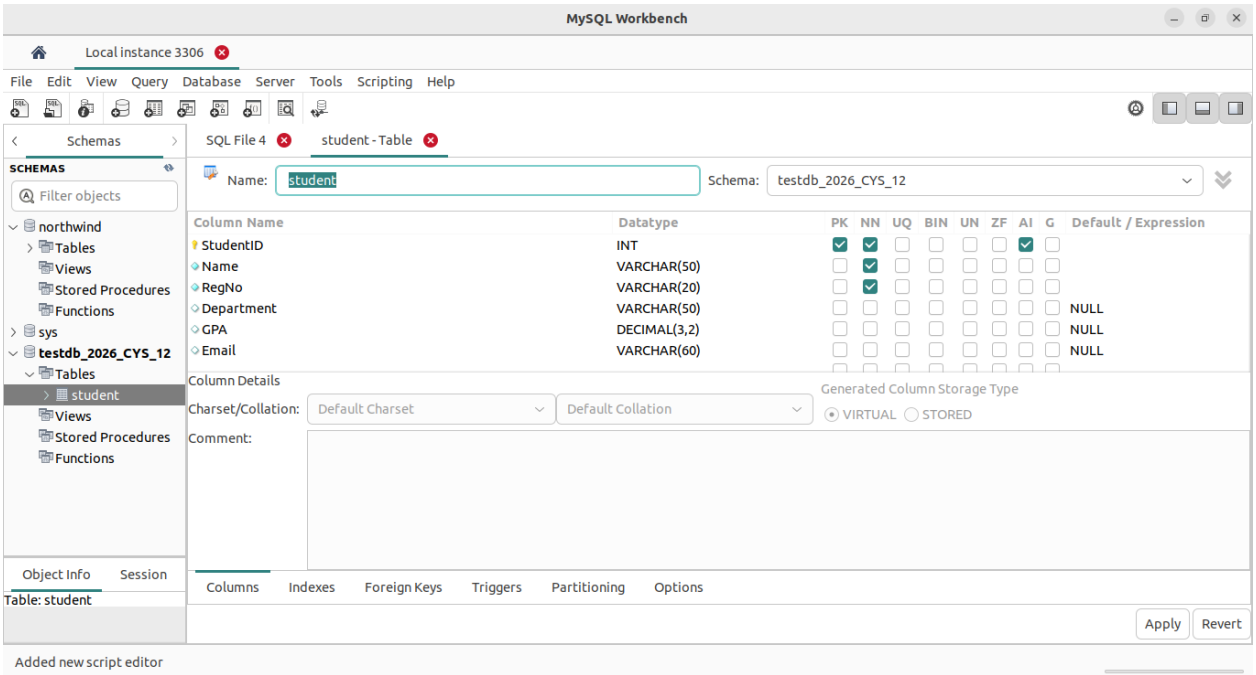

MySQL Workbench
Local instance 3306
File Edit View Query Database Server Tools Scripting Help
Administration Schemas
SCHEMAS
Filter objects
northwind
Tables
customers
employee_privileges
employees
inventory_transaction_types
inventory_transactions
invoices
order_details
order_details_status
orders
orders_status
orders_tax_status
privileges
products
purchase_order_details
Object Info Session
Schema: sys
SQL File 4* northwind*
Open SQL Script
Open
Cancel
Recent
Home
Desktop
Documents
Downloads
Music
Pictures
Videos
home administrator Desktop
Name Size Type Modified
Dump20260922.sql 71.9 kB SQL code 16:53
northwind.sql 20.7 kB SQL code Sun
northwind-data.sql 149.9 kB SQL code Sun
SQL Files (*.sql)
`email_address` VARCHAR(50) NULL DEFAULT NULL,
`job_title` VARCHAR(50) NULL DEFAULT NULL,
`business_phone` VARCHAR(25) NULL DEFAULT NULL,
`home_phone` VARCHAR(25) NULL DEFAULT NULL,
Opening ...

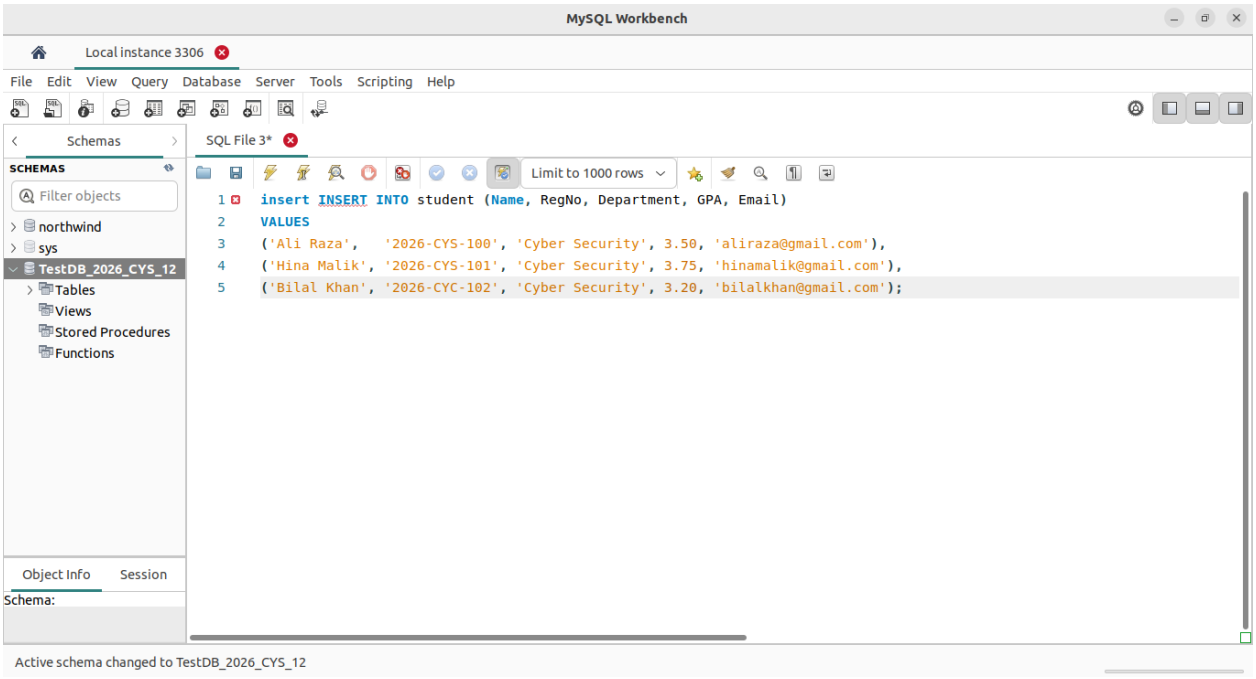

MySQL Workbench
Local instance 3306
File Edit View Query Database Server Tools Scripting Help
Administration Schemas
SCHEMAS
Filter objects
northwind
Tables
customers
Columns
id
company
last_name
first_name
email_address
job_title
business_phone
home_phone
mobile_phone
fax_number
address
city
Object Info Session
Schema: northwind
SQL File 4* northwind* northwind-data
Limit to 1000 rows
Converted from MS Access 2010 Northwind database (northwind.accdb) using
Bullzip MS Access to MySQL Version 5.1.242. http://www.bullzip.com
CHANGES MADE AFTER INITIAL CONVERSION
* column and row names in CamelCase converted to lower_case_with_underscore
* space and slash ("/") in table and column names replaced with _underscore_
* id column names converted to "id"
* foreign key column names converted to xxx_id
* variables of type TIMESTAMP converted to DATETIME to avoid TIMESTAMP
range limitation (1997 - 2038 UTC), and other limitations.
* unique and foreign key checks disabled while loading data
SET @OLD_UNIQUE_CHECKS=@@UNIQUE_CHECKS, UNIQUE_CHECKS=0;
SET @OLD_FOREIGN_KEY_CHECKS=@@FOREIGN_KEY_CHECKS, FOREIGN_KEY_CHECKS=0;
USE `northwind`;
Query Completed

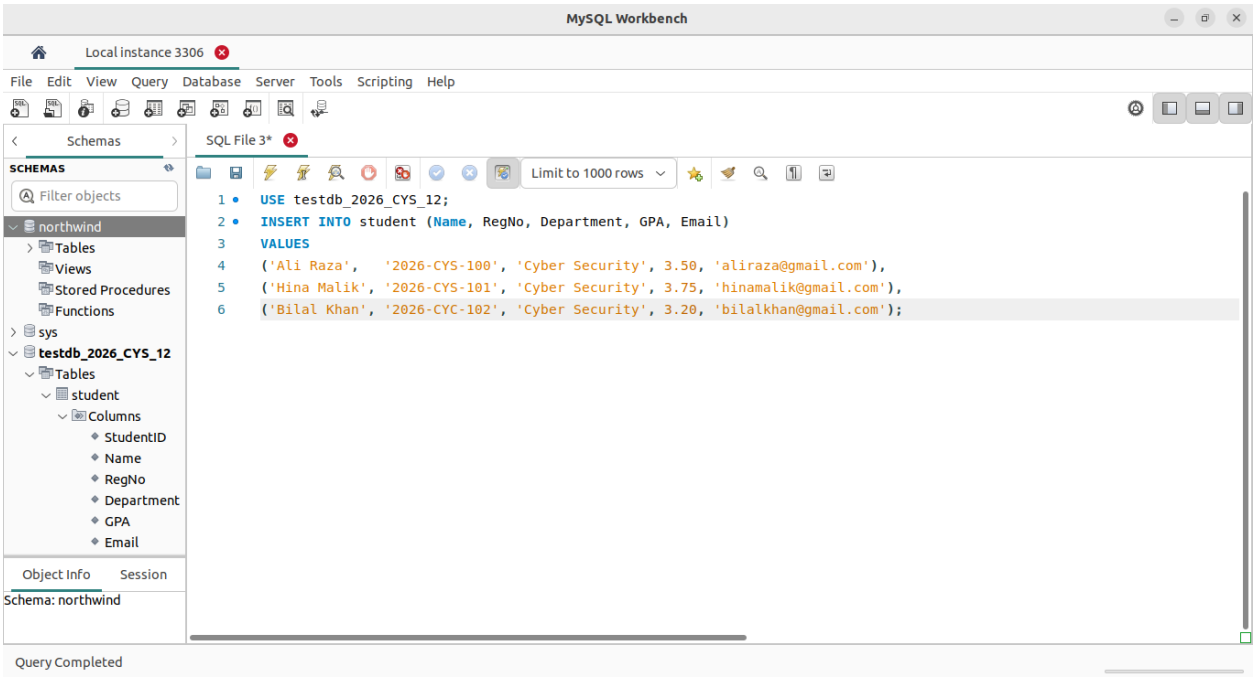


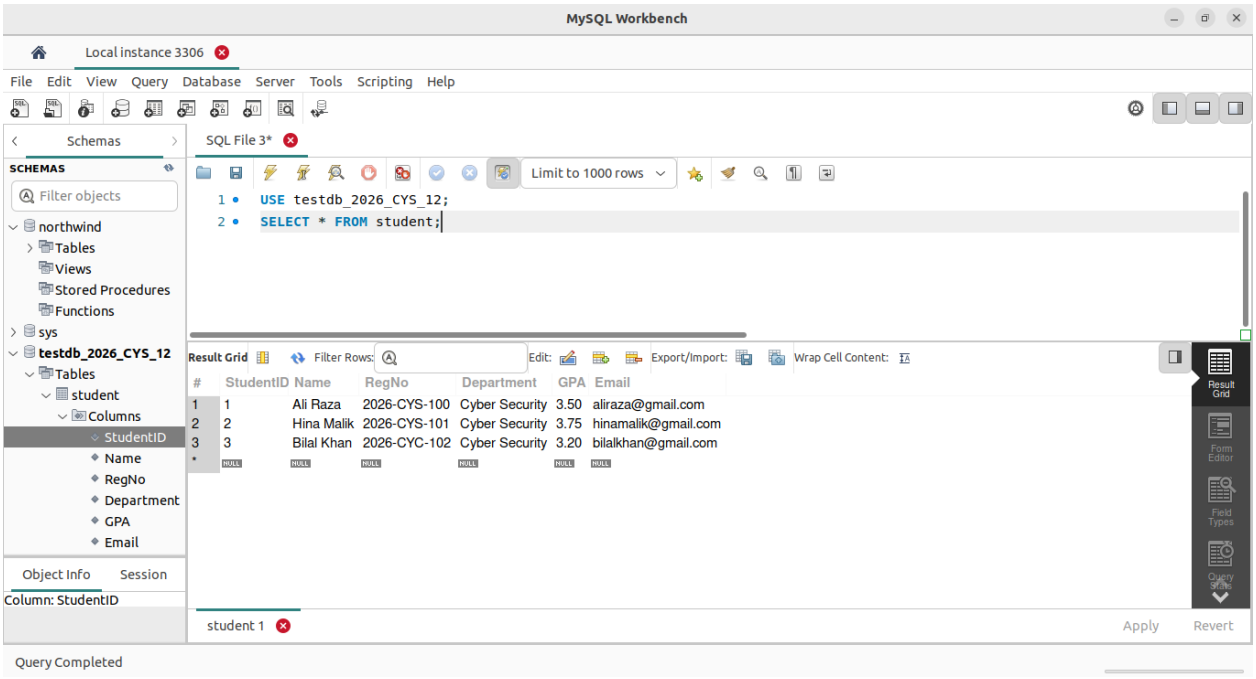
Learning Objective 5: Create a new database to practice the fundamental steps of setting up a database environment

for structured data storage and management.

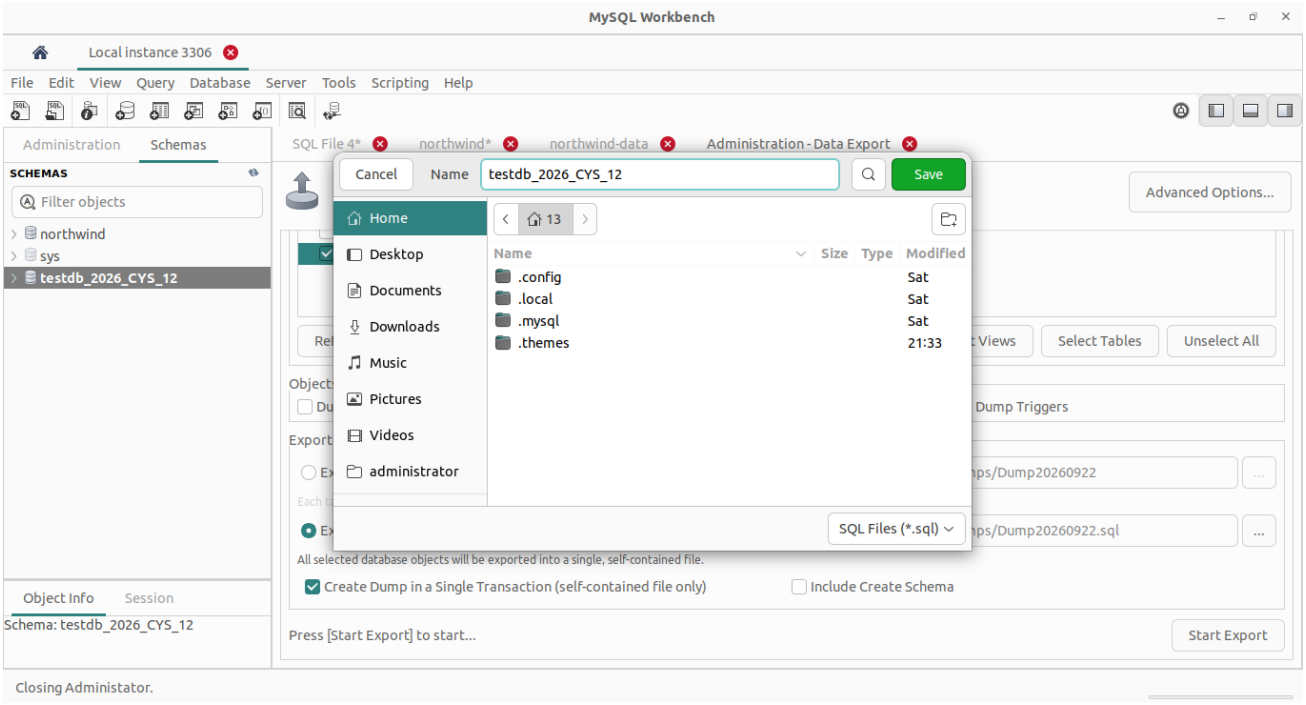


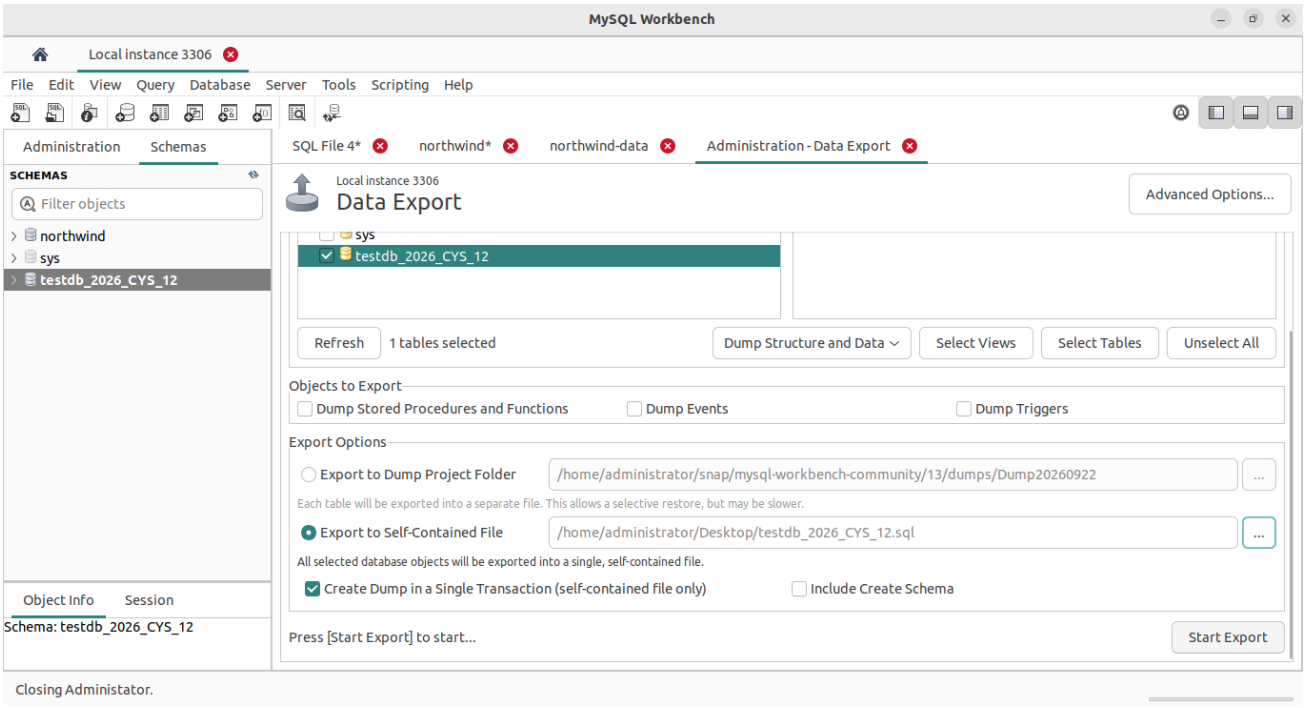

MySQL Workbench
Local instance 3306
File Edit View Query Database Server Tools Scripting Help
Schemas
SCHEMAS
Filter objects
northwind
sys
TestDB_2026_CYS_12
Tables
Views
Stored Procedures
Functions
Object Info
Session
Schema:
SQL File 3*
Limit to 1000 rows
1 insert INSERT INTO student (Name, RegNo, Department, GPA, Email)
2 VALUES
3 ('Ali Raza', '2026-CYS-100', 'Cyber Security', 3.50, 'aliraza@gmail.com'),
4 ('Hina Malik', '2026-CYS-101', 'Cyber Security', 3.75, 'hinamalik@gmail.com'),
5 ('Bilal Khan', '2026-CYC-102', 'Cyber Security', 3.20, 'bilalkhan@gmail.com');
Active schema changed to TestDB_2026_CYS_12


MySQL Workbench
Local instance 3306
File Edit View Query Database Server Tools Scripting Help
Schemas
SCHEMAS
Filter objects
northwind
Tables
Views
Stored Procedures
Functions
sys
testdb_2026_CYS_12
Tables
student
Columns
StudentID
Name
RegNo
Department
GPA
Email
Object Info
Session
Schema: northwind
SQL File 3*
Limit to 1000 rows
1 USE testdb_2026_CYS_12;
2 INSERT INTO student (Name, RegNo, Department, GPA, Email)
3 VALUES
4 ('Ali Raza', '2026-CYS-100', 'Cyber Security', 3.50, 'aliraza@gmail.com'),
5 ('Hina Malik', '2026-CYS-101', 'Cyber Security', 3.75, 'hinamalik@gmail.com'),
6 ('Bilal Khan', '2026-CYC-102', 'Cyber Security', 3.20, 'bilalkhan@gmail.com');
Query Completed



Learning Objective 6: Generate scripts for a database schema to understand how database structures (e.g., tables, constraints) are defined and exported.




MySQL Workbench
Local instance 3306
File Edit View Query Database Server Tools Scripting Help
Administration Schemas
SCHEMAS
Filter objects
northwind
sys
testdb_2026_CYS_12
Object Info Session
Schema: testdb_2026_CYS_12
SQL File 4*
northwind*
northwind-data
Administration - Data Export
Local instance 3306
Data Export
Advanced Options...
sys
testdb_2026_CYS_12
Refresh
1 tables selected
Dump Structure and Data
Select Views
Select Tables
Unselect All
Objects to Export
Dump Stored Procedures and Functions
Dump Events
Dump Triggers
Export Options
Export to Dump Project Folder
/home/administrator/snap/mysql-workbench-community/13/dumps/Dump20260922
Each table will be exported into a separate file. This allows a selective restore, but may be slower.
Export to Self-Contained File
/home/administrator/Desktop/testdb_2026_CYS_12.sql
All selected database objects will be exported into a single, self-contained file.
Create Dump in a Single Transaction (self-contained file only)
Include Create Schema
Press [Start Export] to start...
Start Export
Closing Administator.


Open
testdb_2026_CYS_12.sql
~/Desktop/2026_S_CYS_12
Save
-- MySQL dump 10.13 Distrib 8.0.36, for Linux (x86_64)
--
-- Host: localhost Database: testdb_2026_CYS_12
-- --
-- Server version 8.0.46-0ubuntu0.22.04.4
/*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */;
/*!40101 SET @OLD_CHARACTER_SET_RESULTS=@@CHARACTER_SET_RESULTS */;
/*!40101 SET @OLD_COLLATION_CONNECTION=@@COLLATION_CONNECTION */;
/*!50503 SET NAMES utf8 */;
/*!40103 SET @OLD_TIME_ZONE=@@TIME_ZONE */;
/*!40103 SET TIME_ZONE='+00:00' */;
/*!40014 SET @OLD_UNIQUE_CHECKS=@@UNIQUE_CHECKS, UNIQUE_CHECKS=0 */;
/*!40014 SET @OLD_FOREIGN_KEY_CHECKS=@@FOREIGN_KEY_CHECKS, FOREIGN_KEY_CHECKS=0 */;
/*!40101 SET @OLD_SQL_MODE=@@SQL_MODE, SQL_MODE='NO_AUTO_VALUE_ON_ZERO' */;
/*!40111 SET @OLD_SQL_NOTES=@@SQL_NOTES, SQL_NOTES=0 */;
--
-- Table structure for table `student`
--
DROP TABLE IF EXISTS `student`;
/*!40101 SET @saved_cs_client = @@character_set_client */;
/*!50503 SET character_set_client = utf8mb4 */;
CREATE TABLE `student` (
`StudentID` int NOT NULL AUTO_INCREMENT,
`Name` varchar(50) NOT NULL,
`RegNo` varchar(20) NOT NULL,
`Department` varchar(50) DEFAULT NULL,
`GPA` decimal(3,2) DEFAULT NULL,
`Email` varchar(60) DEFAULT NULL,
PRIMARY KEY (`StudentID`)
) ENGINE=InnoDB AUTO_INCREMENT=4 DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
/*!40101 SET character_set_client = @saved_cs_client */;
--
-- Dumping data for table `student`
SQL
Tab Width: 8
Ln 1, Col 1
INS

Open

Screenshot captured
You can paste the image from the clipboard.

Save

```
--

DROP TABLE IF EXISTS `student`;
/*!40101 SET @saved_cs_client     = @@character_set_client */;
/*!50503 SET character_set_client = utf8mb4 */;
CREATE TABLE `student` (
  `StudentID` int NOT NULL AUTO_INCREMENT,
  `Name` varchar(50) NOT NULL,
  `RegNo` varchar(20) NOT NULL,
  `Department` varchar(50) DEFAULT NULL,
  `GPA` decimal(3,2) DEFAULT NULL,
  `Email` varchar(60) DEFAULT NULL,
  PRIMARY KEY (`StudentID`)
) ENGINE=InnoDB AUTO_INCREMENT=4 DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
/*!40101 SET character_set_client = @saved_cs_client */;

--
-- Dumping data for table `student`
--

LOCK TABLES `student` WRITE;
/*!40000 ALTER TABLE `student` DISABLE KEYS */;
INSERT INTO `student` VALUES (1,'Ali Raza','2026-CYS-100','Cyber Security',3.50,'aliraza@gmail.com'),(2,'Hina Malik','2026-CYS-101','Cyber Security',3.75,'hinamalik@gmail.com'),(3,'Bilal Khan','2026-CYC-102','Cyber Security',3.20,'bilalkhan@gmail.com');
/*!40000 ALTER TABLE `student` ENABLE KEYS */;
UNLOCK TABLES;
/*!40103 SET TIME_ZONE=@OLD_TIME_ZONE */;

/*!40101 SET SQL_MODE=@OLD_SQL_MODE */;
/*!40014 SET FOREIGN_KEY_CHECKS=@OLD_FOREIGN_KEY_CHECKS */;
/*!40014 SET UNIQUE_CHECKS=@OLD_UNIQUE_CHECKS */;
/*!40101 SET CHARACTER_SET_CLIENT=@OLD_CHARACTER_SET_CLIENT */;
/*!40101 SET CHARACTER_SET_RESULTS=@OLD_CHARACTER_SET_RESULTS */;
/*!40101 SET COLLATION_CONNECTION=@OLD_COLLATION_CONNECTION */;
/*!40111 SET SQL_NOTES=@OLD_SQL_NOTES */;

-- Dump completed on 2026-09-22 23:14:28
```

SQL Tab Width: 8 Ln 1, Col 1 INS

MySQL Workbench

Local instance 3306

File Edit View Query Database Server Tools Scripting Help

Administration Schemas

SCHEMAS

Filter objects

- northwind
- sys
- testdb_2026_CYS_12

Object Info Session

Schema: testdb_2026_CYS_12

SQL File 4* northwind* northwind-data Administration - Data Export

Local instance 3306

Data Export

Advanced Options...

Expo Schema

- northwind
- sys
- testdb_2026_CYS_12

Expo Schema Objects

- student

Refresh 1 tables selected Dump Structure Only Select Views Select Tables Unselect All

Objects to Export

Dump Stored Procedures and Functions Dump Events Dump Triggers

Export Options

Export to Dump Project Folder /home/administrator/snap/mysql-workbench-community/13/dumps/Dump20260922 ...

Each table will be exported into a separate file. This allows a selective restore, but may be slower.

Export to Self-Contained File /home/administrator/snap/mysql-workbench-community/13/structure_only.sql ...

All selected database objects will be exported into a single, self-contained file.

Create Dump in a Single Transaction (self-contained file only) Include Create Schema

Closing Administator.

Open ▾ | structure_only.sql — ~/Desktop/2026_S_CYS_12 | Save

```sql
-- MySQL dump 10.13  Distrib 8.0.36, for Linux (x86_64)
--
-- Host: localhost    Database: testdb_2026_CYS_12
-- ------------------------------------------------------
-- Server version 8.0.46-0ubuntu0.22.04.4

/*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */;
/*!40101 SET @OLD_CHARACTER_SET_RESULTS=@@CHARACTER_SET_RESULTS */;
/*!40101 SET @OLD_COLLATION_CONNECTION=@@COLLATION_CONNECTION */;
/*!50503 SET NAMES utf8 */;
/*!40103 SET @OLD_TIME_ZONE=@@TIME_ZONE */;
/*!40103 SET TIME_ZONE='+00:00' */;
/*!40014 SET @OLD_UNIQUE_CHECKS=@@UNIQUE_CHECKS, UNIQUE_CHECKS=0 */;
/*!40014 SET @OLD_FOREIGN_KEY_CHECKS=@@FOREIGN_KEY_CHECKS, FOREIGN_KEY_CHECKS=0 */;
/*!40101 SET @OLD_SQL_MODE=@@SQL_MODE, SQL_MODE='NO_AUTO_VALUE_ON_ZERO' */;
/*!40111 SET @OLD_SQL_NOTES=@@SQL_NOTES, SQL_NOTES=0 */;

--
-- Table structure for table `student`
--

DROP TABLE IF EXISTS `student`;
/*!40101 SET @saved_cs_client     = @@character_set_client */;
/*!50503 SET character_set_client = utf8mb4 */;
CREATE TABLE `student` (
  `StudentID` int NOT NULL AUTO_INCREMENT,
  `Name` varchar(50) NOT NULL,
  `RegNo` varchar(20) NOT NULL,
  `Department` varchar(50) DEFAULT NULL,
  `GPA` decimal(3,2) DEFAULT NULL,
  `Email` varchar(60) DEFAULT NULL,
  PRIMARY KEY (`StudentID`)
) ENGINE=InnoDB AUTO_INCREMENT=4 DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci;
/*!40101 SET character_set_client = @saved_cs_client */;
/*!40103 SET TIME_ZONE=@OLD_TIME_ZONE */;

/*!40101 SET SQL_MODE=@OLD_SQL_MODE */;
```

SQL ▾ | Tab Width: 2 ▾ | Ln 1, Col 1 | INS

MySQL Workbench

Local instance 3306

File Edit View Query Database Server Tools Scripting Help

Administration | Schemas

SCHEMAS

Filter objects

- northwind
- sys
- testdb_2026_CYS_12

Object Info | Session

Schema: testdb_2026_CYS_12

SQL File 4* | northwind* | northwind-data | Administration - Data Export

Advanced Options...

Select Tables | Unselect All

Dump Triggers

Export

ps/Dump20260922

ps/Dump20260922.sql

All selected database objects will be exported into a single, self-contained file.

Create Dump in a Single Transaction (self-contained file only) | Include Create Schema

Press [Start Export] to start...

Start Export

Closing Administator.

Cancel | Name: data_only | Save

Home | Desktop | Documents | Downloads | Music | Pictures | Videos | administrator

home > administrator > Desktop

Name	Size	Type	Modified
2026_S_CYS_12			23:16
northwind.sql	20.7 kB	SQL code	Sun
northwind-data.sql	149.9 kB	SQL code	Sun

SQL Files (*.sql)

Open | data_only.sql | ~/Desktop/2026_S_CYS_12 | Save

```sql
-- MySQL dump 10.13  Distrib 8.0.36, for Linux (x86_64)
--
-- Host: localhost    Database: testdb_2026_CYS_12
-- ------------------------------------------------------
-- Server version       8.0.46-0ubuntu0.22.04.4

/*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */;
/*!40101 SET @OLD_CHARACTER_SET_RESULTS=@@CHARACTER_SET_RESULTS */;
/*!40101 SET @OLD_COLLATION_CONNECTION=@@COLLATION_CONNECTION */;
/*!50503 SET NAMES utf8 */;
/*!40103 SET @OLD_TIME_ZONE=@@TIME_ZONE */;
/*!40103 SET TIME_ZONE='+00:00' */;
/*!40014 SET @OLD_UNIQUE_CHECKS=@@UNIQUE_CHECKS, UNIQUE_CHECKS=0 */;
/*!40014 SET @OLD_FOREIGN_KEY_CHECKS=@@FOREIGN_KEY_CHECKS, FOREIGN_KEY_CHECKS=0 */;
/*!40101 SET @OLD_SQL_MODE=@@SQL_MODE, SQL_MODE='NO_AUTO_VALUE_ON_ZERO' */;
/*!40111 SET @OLD_SQL_NOTES=@@SQL_NOTES, SQL_NOTES=0 */;

--
-- Dumping data for table `student`
--

LOCK TABLES `student` WRITE;
/*!40000 ALTER TABLE `student` DISABLE KEYS */;
INSERT INTO `student` VALUES (1,'Ali Raza','2026-CYS-100','Cyber Security',3.50,'aliraza@gmail.com'),(2,'Hina Malik','2026-CYS-101','Cyber Security',3.75,'hinamalik@gmail.com'),(3,'Bilal Khan','2026-CYC-102','Cyber Security',3.20,'bilalkhan@gmail.com');
/*!40000 ALTER TABLE `student` ENABLE KEYS */;
UNLOCK TABLES;
/*!40103 SET TIME_ZONE=@OLD_TIME_ZONE */;

/*!40101 SET SQL_MODE=@OLD_SQL_MODE */;
/*!40014 SET FOREIGN_KEY_CHECKS=@OLD_FOREIGN_KEY_CHECKS */;
/*!40014 SET UNIQUE_CHECKS=@OLD_UNIQUE_CHECKS */;
/*!40101 SET CHARACTER_SET_CLIENT=@OLD_CHARACTER_SET_CLIENT */;
/*!40101 SET CHARACTER_SET_RESULTS=@OLD_CHARACTER_SET_RESULTS */;
/*!40101 SET COLLATION_CONNECTION=@OLD_COLLATION_CONNECTION */;
/*!40111 SET SQL_NOTES=@OLD_SQL_NOTES */;

```

SQL | Tab Width: 8 | Ln 1, Col 1 | INS